Appendix D.3 — Requirement Traceability and Verification Outcomes

Purpose

This document provides the detailed requirement-to-verification matrix supporting §5.11 of the CurriculumAI dissertation. It records how the 12 functional requirements (FR01–FR12) and seven non-functional requirements (NFR01–NFR07), defined in §3.4 and specified in Appendix D.2, were addressed through implemented system components and selected verification tests.

Table D.3 establishes traceability across four elements: requirement identifier, implemented component, executed test evidence and observed outcome. The test identifiers correspond to the unit (U), integration (I), functional (FT), error/edge-case (E), browser (B) and full-system (S) verification cases documented in Appendix A.

ID	Implemented component	Test evidence	Observed outcome
FR01	QA/ context	FT02-FT03, B02-B03, S01	Guest and registered chat workflows verified
FR02	Retrieval/ answer routing	U04, I02, I08, E04	Retrieval ranking and routing checks passed
FR03	Tier-1 /citations	U03, FT04, B04, S01	Citation validation and live display verified
FR04	Tier-2 fallback	U05, FT05, B05, S02	Fallback verified, including live workflow
FR05	Error recovery	U05, E04, E06, E08	Controlled recovery and error handling verified
FR06	Student accounts	FT01, B01, B08	Account workflows passed

FR07	Private history	I04, FT03, B03, S01	Ownership/storage and reopening verified
FR08	Source ingestion	I01, I07, E01-E03, B06, S03	Ingestion, retry and live indexing verified
FR09	Source deletion	I05, B04	Historical citation snapshots preserved
FR10	Admin reporting	B06, E07	Usage display and access controls checked
FR11	Tier-model setting	U06, I06, B07	Independent settings and persistence verified
FR12	Admin account	FT06, E07	Account and permissions checks passed
NFR01	Interface	B01-B08, S01-S03	Interaction checks passed
NFR02	Source transparency	U03, FT04-FT05, B04-B05, S01-S02	Tier labels and source presentation verified
NFR03	Ingestion reliability	I07, E03, S03	Failure isolation and ingestion workflow verified
NFR04	Privacy and security	I04, E07	Selected safeguards verified
NFR05	Maintainability	U01-U06, I01-I08	Structural testability evidence only
NFR06	Source extensibility	I01, B06, S03	Supported document workflows verified
NFR07	Model persistence	U06, I06, B07	Settings persisted across restart; request consistency verified

Table D.3: Requirement Traceability and Verification Outcomes

Interpretation and Scope of Verification

The matrix documents verification of the specified behaviours under the executed test conditions. The three live full-system tests (S01–S03) provide evidence of integrated Tier-1 answering, Tier-2 fallback and source ingestion workflows.

The outcomes do not constitute unrestricted assurance of all system qualities. In particular, NFR04 is supported by verification of selected privacy and security safeguards, while NFR05 is supported by evidence of modularity and structural testability rather than a comprehensive maintainability assessment.

Detailed test procedures and results are available in Appendix A. The comparative answer-quality and student evaluations are reported separately in Chapter 6.